OPEN AGENT HACKATHON 2026 · TRACK 2 — THE AGENT THAT CAN EXPLAIN WHY

RecallWhy

Agents that tell a retailer which recalled products are on its shelves, which lots, which stores, how many customers to call — **and why they believe it.**

Zetaris federated SQL · Meterless H-MEM + World Model · NVIDIA Nemotron · built in Cursor

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github.com/mikey92/recallwhy

1 · PROBLEM AND USERS

517 federal recall notices since July. Each one is a manual cross-check.

A regional retailer's **product-safety / QA manager** reads every CPSC and FDA notice and must answer:

- Do we sell this — or something that only looks like it?
- Which lots? Which stores still hold them?
- How many customers bought it since the recall began?
- Who supplied it?

184 + 333

CPSC recalls and FDA food enforcement reports, 1 Jul – 21 Oct 2026, read live by RecallWhy

The answers live in three places that don't talk to each other: the **recall feeds**, the **ERP** and the **supplier portal**. A wrong "not ours" leaves a contaminated lot on the shelf; a wrong "ours" pulls good stock.

2 · SOLUTION

Four agents investigate every recall and return a verdict with its evidence.

Scout

Reads both live feeds and the item master through Zetaris; one federated UPC sweep; opens a case per recall that touches the catalog.

Investigator

UPC / model rules, Nemotron judge for text matches, lot codes, prior decisions from H-MEM, exposure by store and lot with federated SQL.

Auditor

Can block. Re-checks the live notice, quotes and lots verbatim, re-derives every number with its own SQL, checks each brief sentence. ≤2 revisions, then a human.

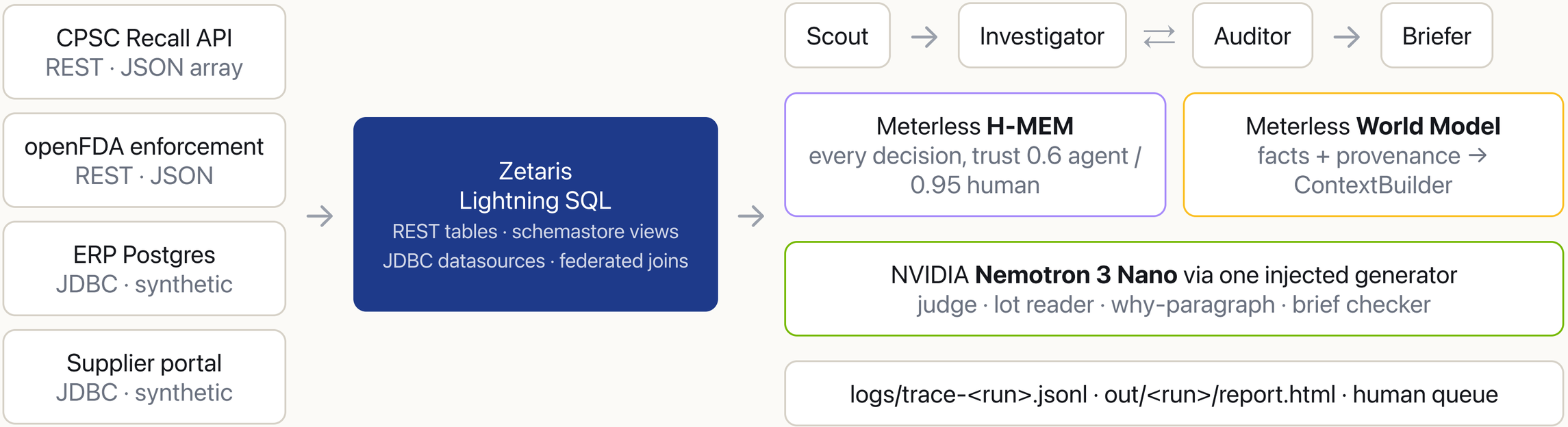
Briefer

Writes only from the World Model context: numbered evidence, each with the SQL or model call behind it.

ACT NOW · NOTIFY · NO EXPOSURE · NOT OURS · NEEDS HUMAN

3 · ARCHITECTURE

Every agent reads data only through Zetaris.



4 · STACK

Small, typed, runnable from a clean clone.

LAYER	WHAT
Data access	Zetaris Lightning SQL over its REST API (`/api/proxy/sql-editor/sqls/run-query`)
Agents	TypeScript on Node 22, no agent framework — four modules, one orchestrator, one trace
Memory & state	Meterless H-MEM and World Model reference engines (Apache-2.0), mounted in `src/engines/`
Model	NVIDIA **nemotron-3-nano** (30B-A3B) via Ollama; any OpenAI-compatible endpoint by env var
Retailer data	Postgres 16 in Docker — synthetic ERP + supplier portal (seeded generator)
No-credential mode	`SAMPLE_MODE=true` : same agents and SQL on an in-process DuckDB stand-in, labelled SAMPLE
Development	Cursor (agent scaffolding, generator, SQL layer, tests) · 37 tests · `npm run view` live trace viewer

5 · HOW ZETARIS AND METERLESS ARE USED

Load-bearing, not bolted on.

Zetaris

- 2 live Lightning REST tables → schemastore views
- 2 JDBC datasources (ERP, supplier portal)
- Scout: **REST × JDBC** join — live FDA notice digits ⋈ ERP UPCs
- Investigator: ship notices ⋈ suppliers ⋈ ERP item master
- Auditor: re-queries the live notice and re-aggregates every number itself
- Setup is code: `npm run setup:zetaris`

Meterless

- **H-MEM**: every adjudication stored with trust; human decisions (0.95) are followed on the next run; audited agent decisions are reused without a model call; trust ledger logs reads/writes
- **World Model**: recalls, cases, findings as facts; every write carries provenance (agent, SQL id, model call)
- **ContextBuilder** is the Briefer's only input — it cannot cite what the team didn't establish

The Auditor doesn't trust anyone — including the model.

- **Re-derives every number in Zetaris** with differently written SQL; a mismatch blocks the brief.
- **Re-reads the live notice** (REST table) to confirm a UPC match, instead of trusting the parser.
- **"Cheese ≠ pepperoni"**: a quote can be copied from the notice and still describe another variety — every product word of our item must appear in the notice. Live run: blocked, re-judged, ruled *not ours*.
- **Lot codes the model reads must appear verbatim** in `code_info`; invented lots are dropped.
- **Brief sentences are checked** against the evidence they cite — it caught a draft that invented stores in Kentucky and Ohio.
- **Human decisions change the next run** through H-MEM, and the brief cites them.

7 · PRODUCT

> Lior brand Cinnamon Ground Seasoning; 90g transparent plastic container; UPC 794711005484.

WHY

The recall notice confirms that Galil Importing Corp is recalling Lior brand Cinnamon Ground Seasoning (UPC 794711005484) due to elevated lead levels E2 E3, and our UPC 794711005484 matches the recalled product E8. Our supplier S024 shipped lots GAP11304, which are explicitly listed in the recall E9, and 284 units remain on hand across seven stores E12 E11, requiring immediate action E17.

WHAT WE MATCHED

- `NW-100093` LIOR BRAND CINNAMON GROUND SEASONING; 90G — **same** (rule, upc): our UPC 794711005484 is printed in the notice

ACTIONS

- Pull lots GAP11304 from 7 store(s): S020 (55), S004 (54), S005 (47), S009 (41), S016 (41), S003 (36) E12
- Send recall notice to loyalty customers who bought affected lots since 2026-09-24: at most 33 E14
- Contact Supplier S024 (synthetic) for credit and replacement E15

CAVEATS

- Retailer data is synthetic (Northwind Mercantile, generated for this demo).
- Stores in OR, WA, WI hold stock although the notice's distribution list does not name those states — product can travel through distributors, so this is not a clearance.
- Customer count sums weekly distinct counts, so it is an upper bound.

EVIDENCE

- E1 notice_firm: "Galil Importing Corp" — *zetaris q-7261132e81 by Scout*
- E2 notice_product_text: "Lior brand Cinnamon Ground Seasoning; 90g transparent plastic container; UPC 794711005484." — *zetaris q-7261132e81 by Scout*
- E3 notice_reason: "Product contains elevated levels of lead." — *zetaris q-7261132e81 by Scout*
- E4 notice_recall_date: "2026-09-24" — *zetaris q-7261132e81 by Scout*
- E5 notice_distribution_states: ["FL","IL","NJ","NY","PA","TX"] — *zetaris q-7261132e81 by Scout*
- E6 notice_code_info: "GAP11304 printed on the side of the container under the ingredient line." — *zetaris q-7261132e81 by Scout*
- E7 notice_upcs: ["794711005484"] — *zetaris q-7261132e81 by Scout*
- E8 match_decision: "NW-100093 LIOR BRAND CINNAMON GROUND SEASONING; 90G: same — our UPC 794711005484 is printed in the notice" — *rule by Investigator*
- E9 our_lots_named_in_notice: ["GAP11304"] — *zetaris q-223c248423 by Investigator*
- E10 lot_rule: "listed-lots" — *agent by Investigator*
- E11 our_stores_holding_affected_lots: 7 — *zetaris q-223c248423 by Investigator*
- E12 our_units_on_hand_in_affected_lots: 284 — *zetaris q-223c248423 by Investigator*
- E13 our_units_sold_since_recall_date: 63 — *zetaris q-34849ae57d by Investigator*
- E14 our_loyalty_customers_who_bought_affected_lots: 33 — *zetaris q-34849ae57d by Investigator*
- E15 our_supplier_for_this_sku: "Supplier S024 (synthetic)" — *zetaris q-9852871a99 by Investigator*

Investigator ollama *["GAP11304"]*

Investigator SQL q- rows, 393 ms

Investigator SQL q- rows, 319 ms

Investigator SQL q- rows, 114 ms

Investigator → Audi stores, 63 sold since

Auditor SQL q-8e4 1402-2026 — 1 rows

Auditor SQL q-700 ms

Auditor SQL q-28c 188 ms

Auditor SQL q-bd53 us — 1 rows, 120 ms

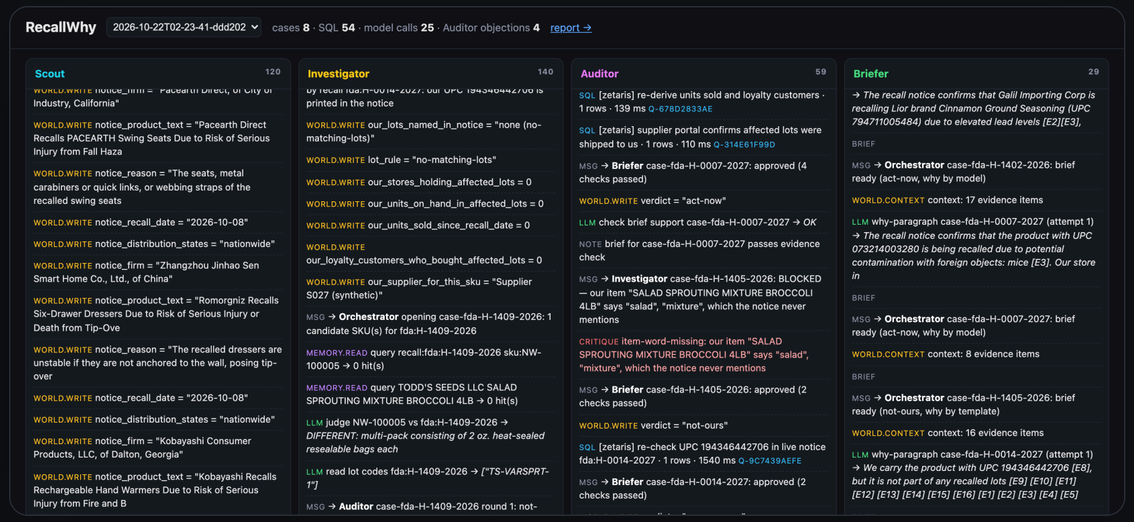
Auditor → Briefer: c

Investigator H-MEN

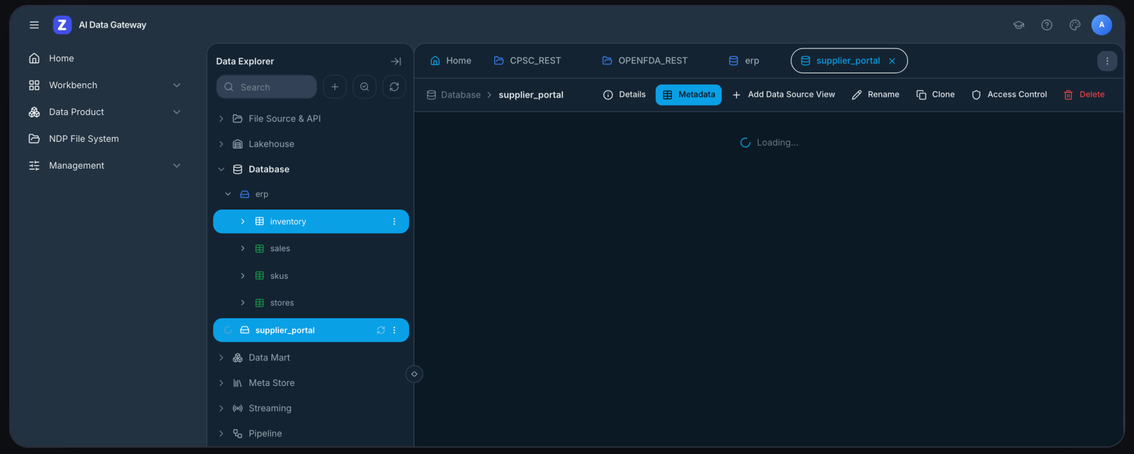
Briefer ollama:nemo → *The recall notice Ground Seasoning (*

Auditor ollama:nem

Brief: verdict, why (cited), actions, caveats, evidence with SQL ids — and the agent trace that produced it.



Live viewer: the four agents' events as they happen, from the trace file.



Zetaris: REST feeds and JDBC sources registered by `setup:zetaris`.

8 · RESULTS

The judge proposes, the Auditor disposes: 207 false alarms → 7, nothing dismissed.

JUDGE MODEL	CAUGHT (OF 194)	WRONGLY DISMISSED	FALSE ALARMS (OF 246)	TO HUMAN
none — rules only	103	0	0	337
NVIDIA Nemotron 3 Nano 30B	**189**	**0**	**7**	**35**
NVIDIA Nemotron 3 Nano 4B	175	18	5	7
Gemma 4 E4B	178	14	5	5

Nemotron 30B's first answers on text matches: **207 false alarms**. After the Auditor's checks and revisions: **7**. The Auditor can prune false matches but cannot recover a dismissal — so the judge must lean toward SAME, and the 30B does.

Real Investigator ⇄ Auditor loop on every candidate pair from 517 recalls (sample mode, DuckDB stand-in). A pair is "covered" if the SKU was generated from that recall or its UPC is printed in the notice. Retailer data is synthetic.

9 · NEXT

From demo to a safety desk.

- More feeds through Zetaris: USDA FSIS, NHTSA equipment (car seats), Health Canada
- Lot capture from receiving scans instead of the item master
- Customer notification through the loyalty system — counts today, messages tomorrow
- Hosted Nemotron through the Meterless runtime for shared deployments
- Markovian engine for daily long-running sweeps
- Review queue UI with one-click decide → H-MEM

Limits today: synthetic retailer data; lexical Auditor checks; CPSC rarely prints UPCs; judge runs on a local Mac.